Informatique et systèmes de communication ISC
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

Signal and information (SigInf)

204.1

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Nota bene on DFT

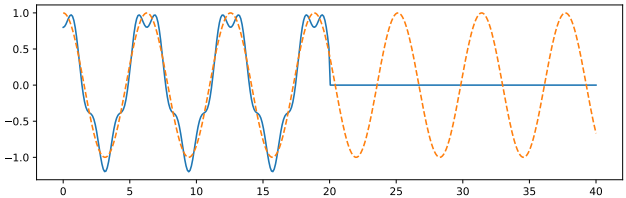
- What really matters in the DFT is the module of the complex values.
- If the goal is to go back to time domain, you need to preserve the (anti)symmetries in frequency domain.

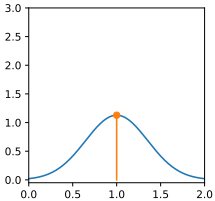
Windowing and filters

- Windows and filters are post-processing steps we apply on signals.
- They correspond to very similar concepts.
- Windows are transforms that we apply to the time domain.
- Filters are transforms that we apply to the frequency domain.

Motivation

We have seen that increasing T (total observation time window, or measurement interval) should **sharpen** the DFT.





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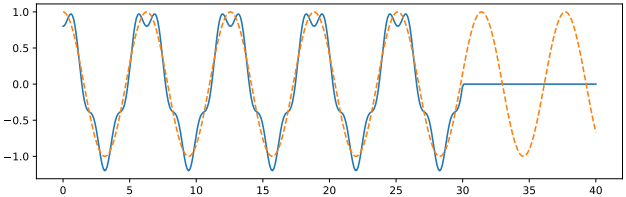

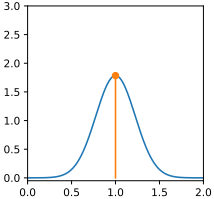

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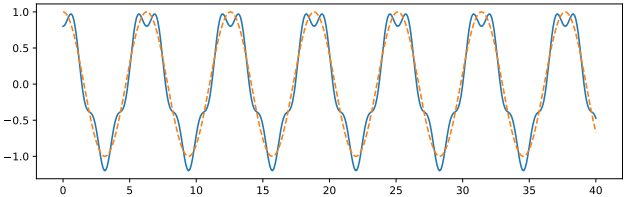

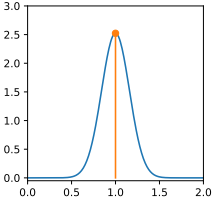

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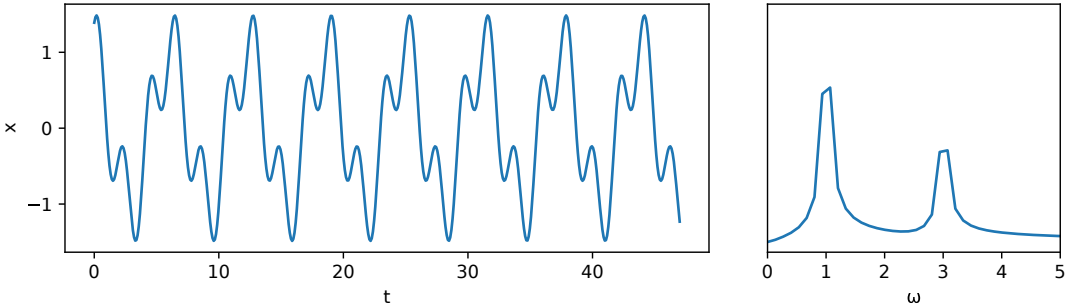

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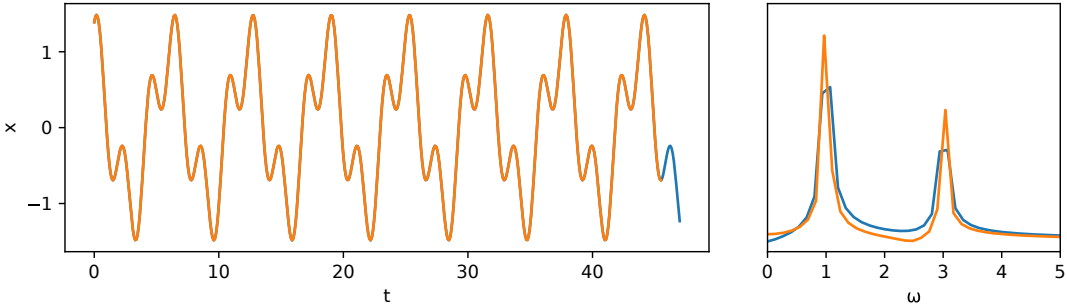

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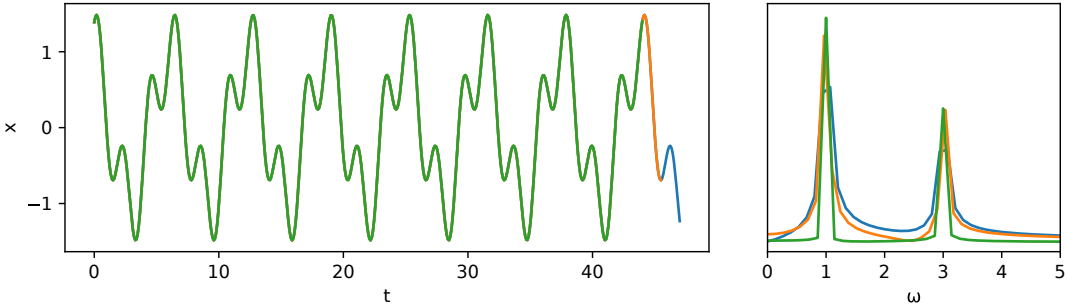

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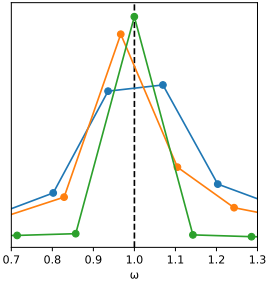

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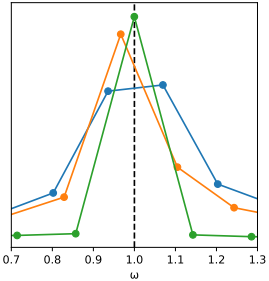

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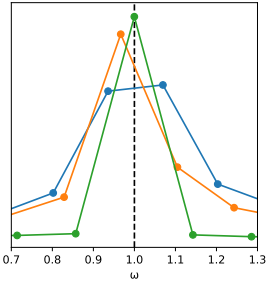

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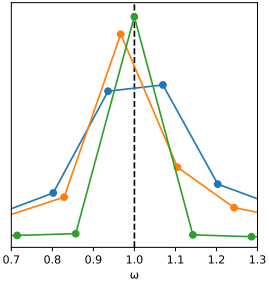

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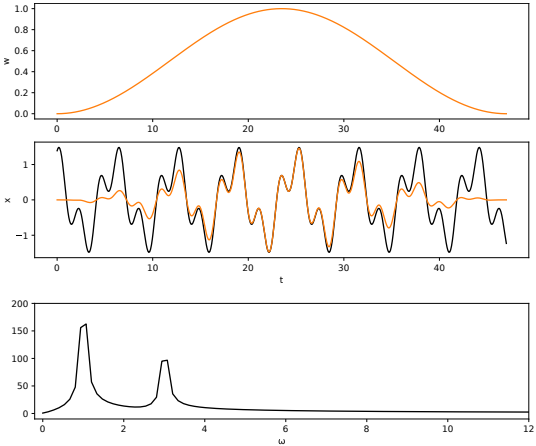
How could we fix that?

What is a window?

We enforce periodicity of the signal over the measurement time.

Take a smooth function that (almost) vanishes at both ends of the measurement interval T.



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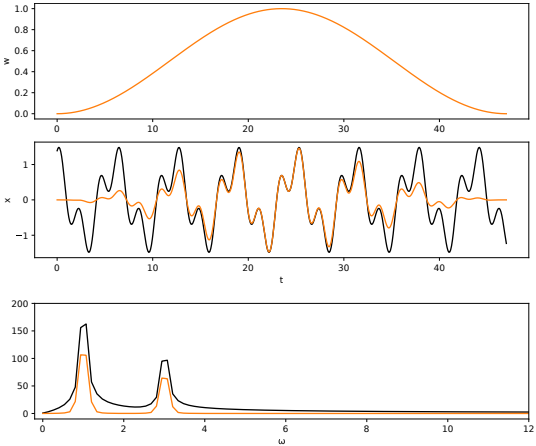

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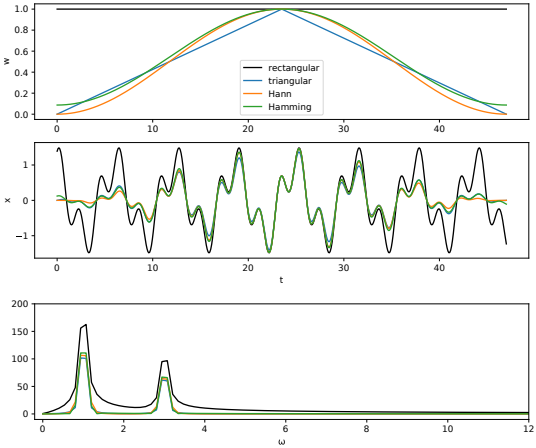
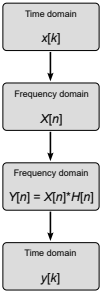


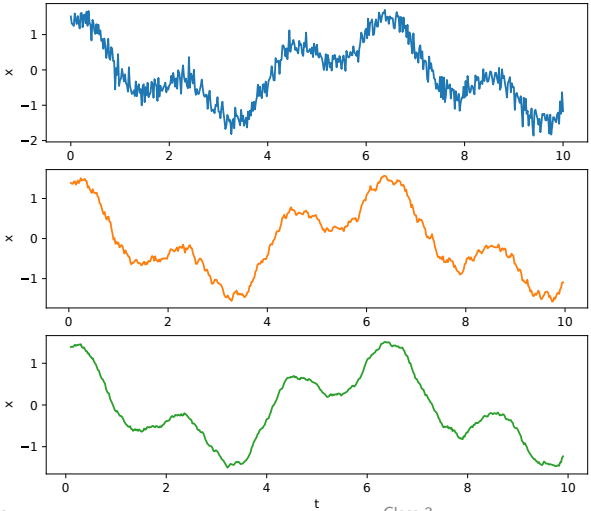
Filtering - definition

A filter is an operation that modifies a signal by preserving desired frequency components and attenuating unwanted ones.



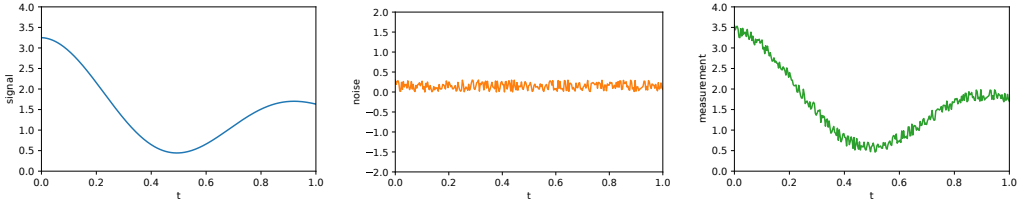
$$x[k] \xrightarrow{\text{FFT}} X[n] \xrightarrow{\times H[n]} Y[n] = X[n]H[n] \xrightarrow{\text{IFFT}} y[k]$$

Smoothing of a signal



Reminder from last class

The **noise** of a signal is the **discrepancy between the measured value and the real value** of the signal.



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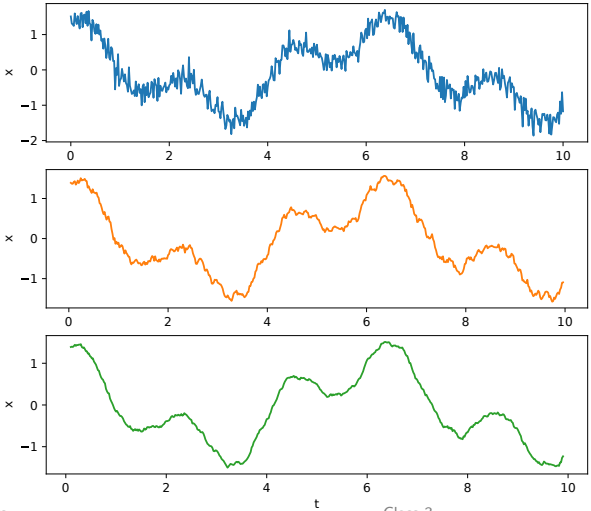

Sliding average

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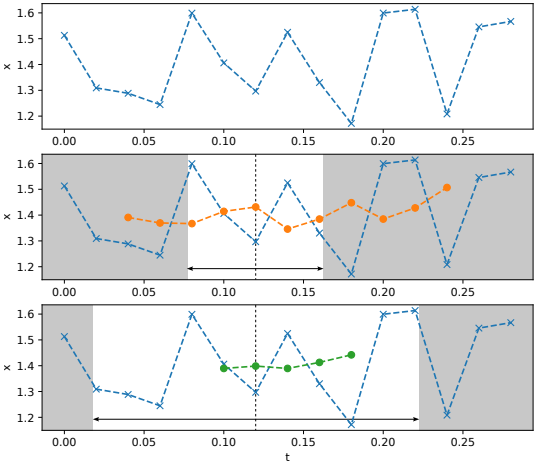
$$\bar{x}[k] = \frac{1}{S} \sum_{j=k-S/2}^{k+S/2} x[j] \,.$$



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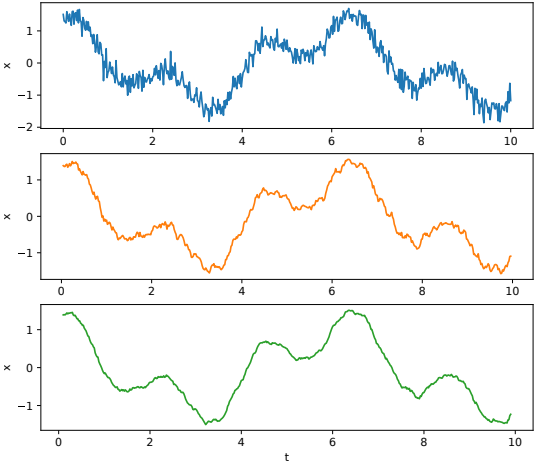
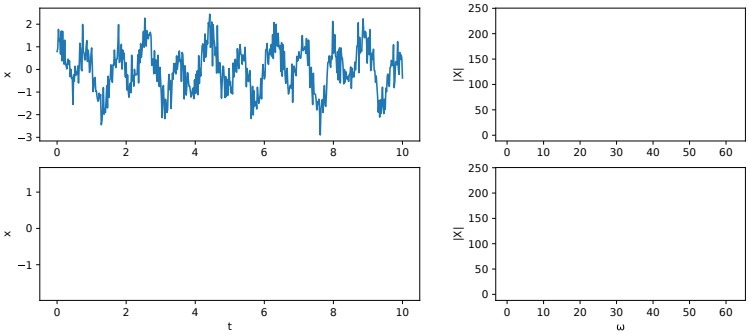


An other way...

Let us look at the FFT of the signal



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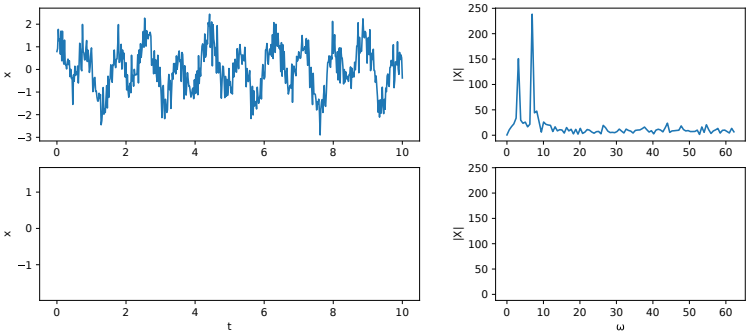

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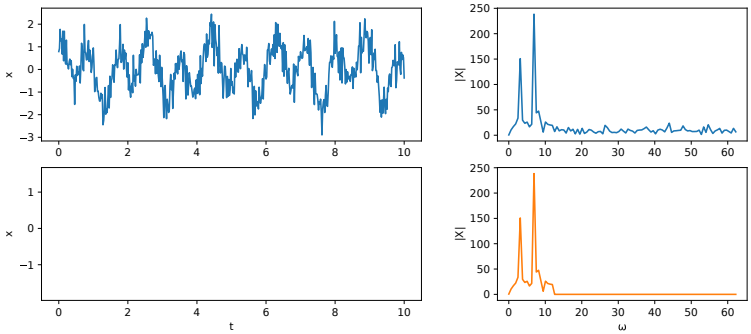

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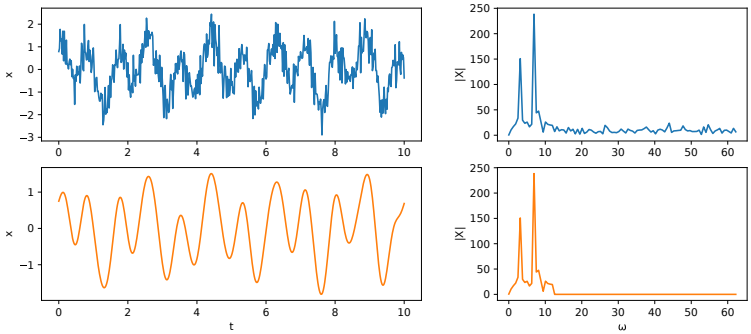
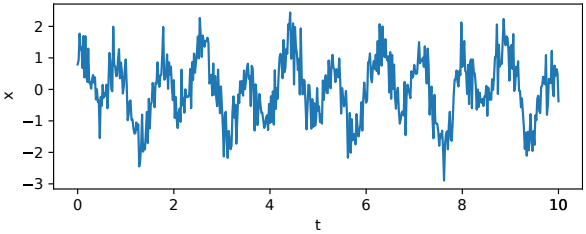
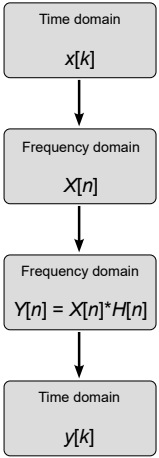


What is a filter? - hands on the example

Same concept as windowing, but in frequency domain.





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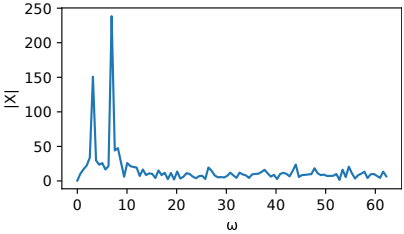

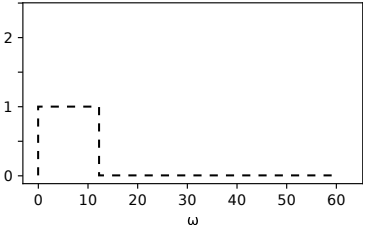

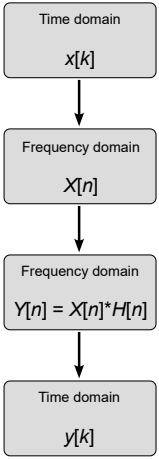

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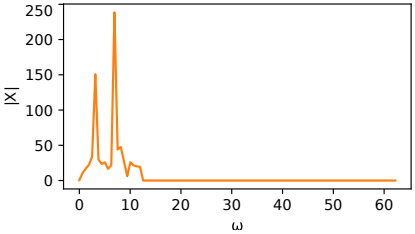

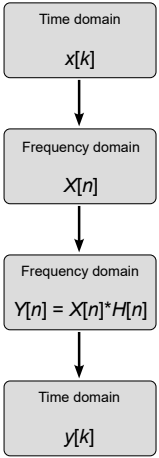

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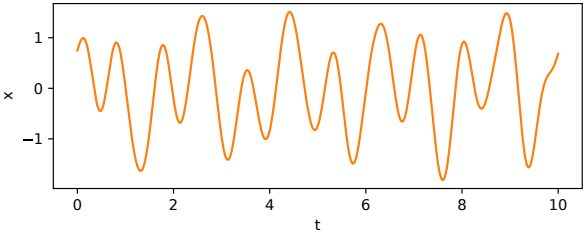
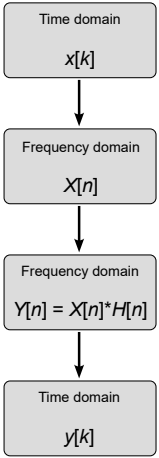
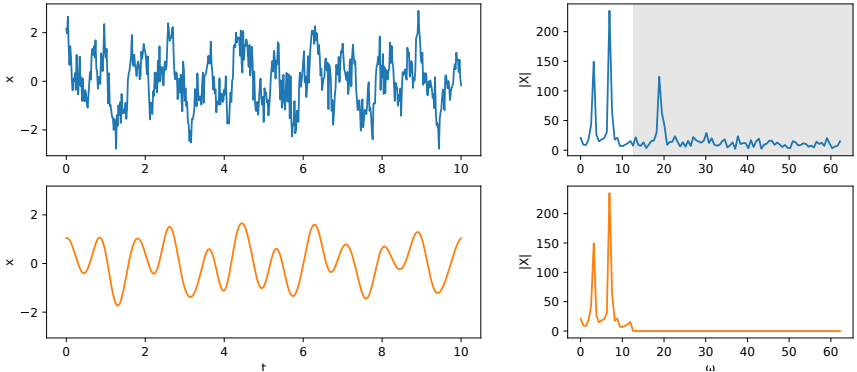




Different types of filters: low-pass filter



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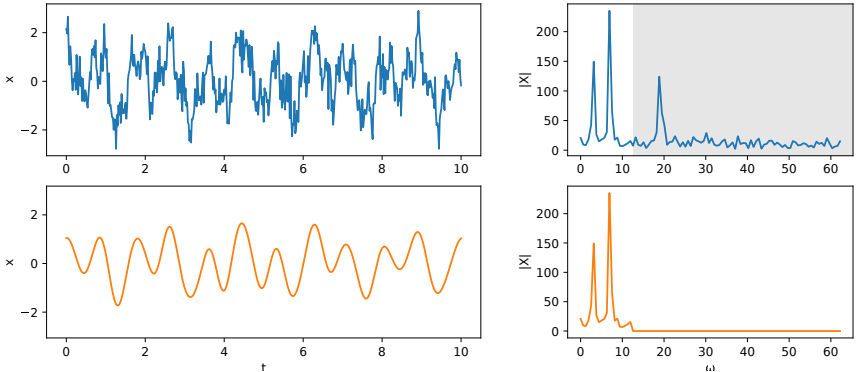
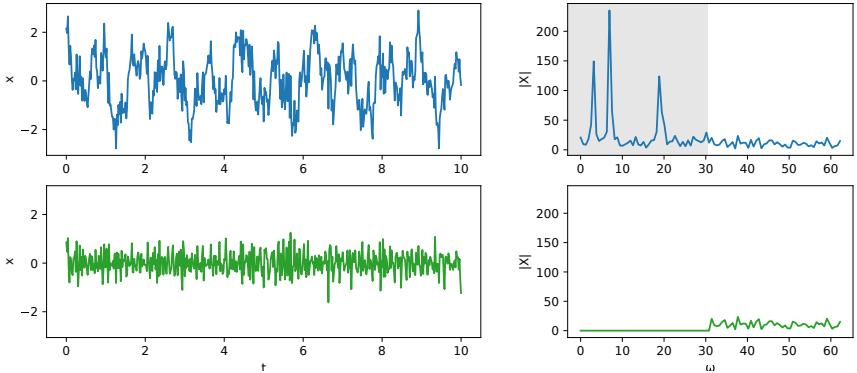


Example: Shock absorber of a car.

Different types of filters: high-pass filter



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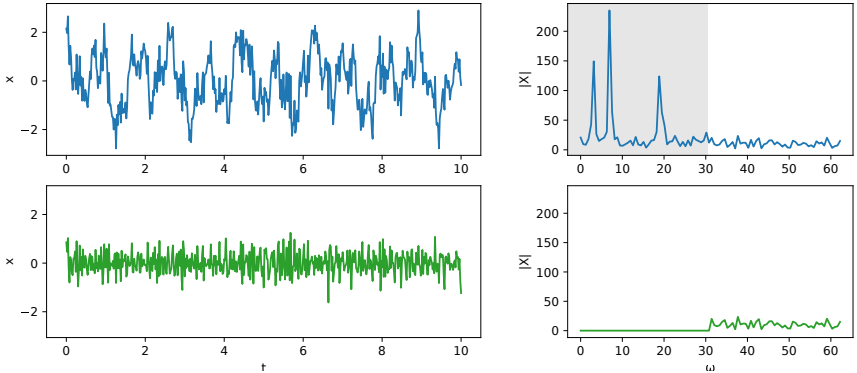
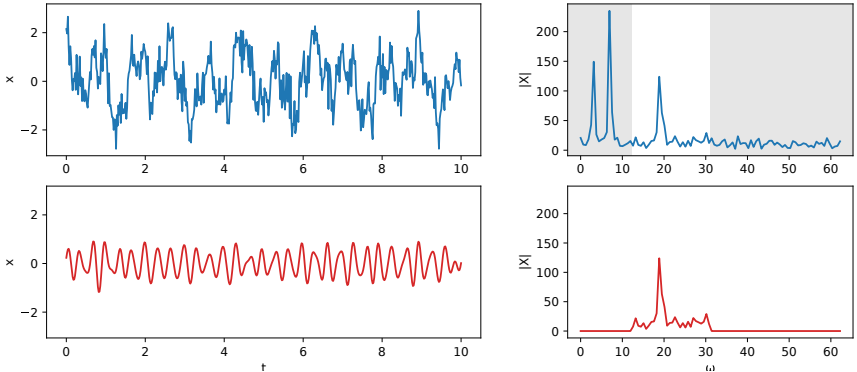


Example: Signal-to-noise ration (SNR), signal sharpening.
White noise contributes to all frequencies

High-pass filter on an image (source: wikipedia)

Different types of filters: band-pass filter



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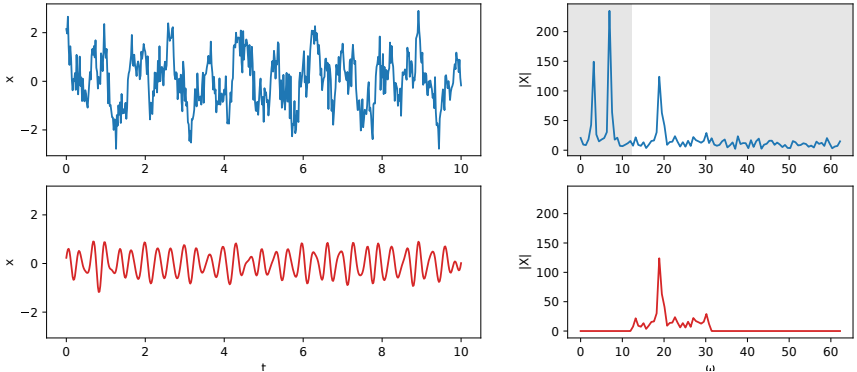
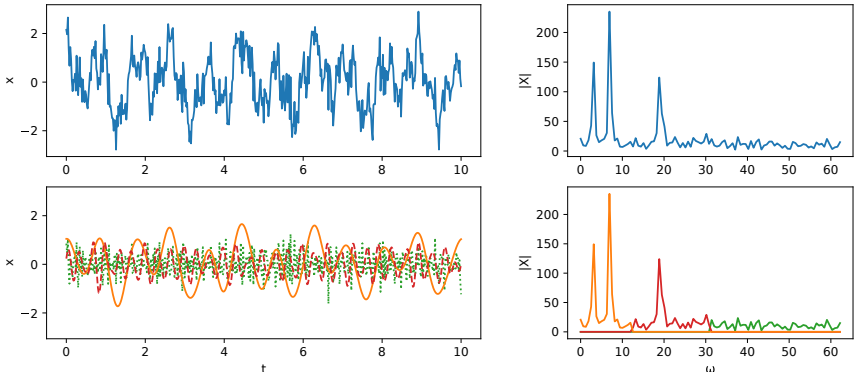


Example: Radio frequency selecter.

Recap



Was the sliding average a filter?

Answer: yes!

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The sliding average is the **convolution** of two signals (h is called a rectangular **kernel**):

$$x[k] = \frac{1}{S} \sum_{j=k-S/2}^{k+S/2} x[j] \qquad \text{and} \qquad h[k] = \begin{cases} \frac{1}{S}, & 0 \leq k \leq S-1, \\ 0, & \text{otherwise.} \end{cases}$$

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The **convolution** of $x[k]$ and $h[k]$ is

$$x[k] * h[k] = \sum_{n=-\infty}^{+\infty} x[n]h[k-n]\,.$$

Convolution and Fourier Transform

Convolution in time becomes multiplication in frequency:

$$y[k] = (x * h)[k] \quad \overset{\text{DFT}}{\Longrightarrow} \quad Y[n] = X[n]H[n]$$

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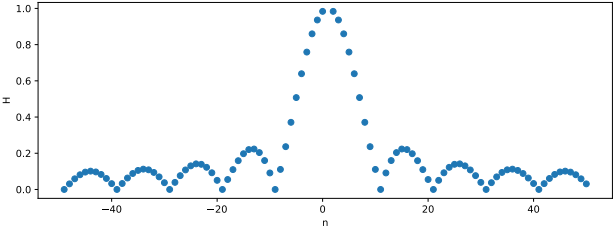
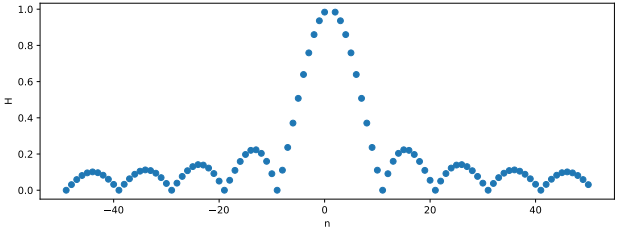


Effect of the averaging length S

- Smaller S preserves more details but removes less noise.
- Larger S gives stronger smoothing in time.
- Larger S makes the low-pass effect stronger in frequency.